

# “Do you get déjà vu”: Persuasiveness Effects of Communicating with an Avatar of Similar Appearance in Social Virtual Reality

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Figure 1: We asked each participant (a) to create a self-representation (b) for use in the VR environment. Their interlocutors' avatars could be of (c) identical appearance, (d) moderately similar appearance, or (e) dissimilar appearance

## ABSTRACT

The similarity effect is a phenomenon whereby people tend to be more influenced by others who resemble them in appearance. It has been found to have many positive impacts, including building trust and promoting collaboration performance. While empirical evidence confirmed that similarity effect operates on screen-based communication platforms, it remains unclear how this phenomenon impacts user perceptions and others' persuasiveness in immersive virtual reality (VR) environments. In this study, we adopted a mixed-methods approach to investigating how interaction with avatars of similar appearance to one's own avatar influences conversations. We operationalized similarity as identity, moderate similarity, and dissimilarity. Avatars of moderate similarity were found to have the greatest persuasiveness; however, in both identity and moderate similarity conditions, participants felt it was easier to communicate with and lower eeriness rating to avatars than in the dissimilarity condition. We conclude that similarity effect could be leveraged to support persuasiveness in VR-based communication.

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## CCS CONCEPTS

• Human-centered computing → Empirical studies in collaborative and social computing.

## KEYWORDS

Computer Mediated Communication, Social Virtual Reality, Avatar, Appearance, Similarity Effect

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## 1 INTRODUCTION

People tend to be more influenced by those who look like them than by dissimilar others [11, 46]. This phenomenon, known as the *similarity effect*, has been studied in computer-mediated communication (CMC) in contexts such as collaboration [21], decision games [14], and brainstorming [10]. Research has suggested that similar appearance in screen-based CMC environments can influence perceptions of credibility [29], trustworthiness [12, 14], attractiveness [56], group identification [10, 40], and persuasiveness [50], as well

as their objective collaboration performance [10, 56], trustworthiness, attractiveness, group identification and persuasiveness, as well as objective collaboration performance. With virtual reality (VR) becoming a promising platform for socializing and collaboration [19], it's important to understand how existing social-interaction theories apply to VR.

In VR, people create digital representations of themselves known as *avatars* and interact with other users in a digital space [25, 39]. Social VR, a subtype of VR, is specifically used for socializing [19]. Social VR allows for more complex relationships between self-representations and other users' avatars than other platforms, due to full-body tracking, the flexibility of avatar appearance, and embodied interaction. Previous studies have shown that appearance plays an important role in shaping people's perceptions [19, 25], task performance [61], and interaction experience [2, 19, 51] in VR. However, it is unknown if the similarity effect applies to avatars' appearance in a VR-based CMC platform.

To investigate the role appearance similarity plays in an immersive conversation, we recruited 25 participants and conducted a mixed-methods study using a VR-based CMC platform. We instructed each of our participants to create an avatar to represent themselves, and then to use it to communicate with a researcher-controlled avatar whose appearance was identical to, moderately like, or unlike the participant's self-representation. We administered questionnaires and conducted semi-structured interviews to examine how these social VR users perceived these different levels of appearance similarity, and applied quantitative measures to investigate how it influenced the researcher-controlled avatars' persuasiveness.

This paper makes two contributions to the HCI research community. First, to the best of our knowledge, it is the first empirical exploration of how people's avatars' interactions with other users' avatars in social VR are affected by appearance similarities of various levels. Second, it provides novel insights into how people in social VR communicate with an avatar of similar appearance, both objectively (e.g., persuasiveness) and subjectively (e.g., intimacy and fluency). More specifically, among all three conditions, people in the moderate similarity condition were most likely to be persuaded to change their initial choices; and, in both that condition and the identical-appearance condition, they reported feeling 1) that it was easier to communicate, 2) that the conversation was more intimate, and 3) that the other avatar was less eerie, as compared to their counterparts in the dissimilar-appearance condition.

These contributions could have important implications regarding the relationship between self-representation and other users' avatars and different levels of appearance similarity when designing future social VR applications.

## 2 RELATED WORK

In this section, we summarize the current state of research on avatars, and how our work builds upon it.

### 2.1 Impacts of Avatars' Appearance

**2.1.1 Self-representations' Appearance.** Previous studies have shown that the *Proteus effect* plays a crucial role in self-representation in virtual environments [48, 62]. It refers to how

the visual appearance of self-representations can affect behavior, attitude, and cognition [9, 31, 32, 34]. For example, embodying a childlike self-representation can lead to overestimating object sizes or adopting more childish behavior [8]. Users' behaviors are also affected when their self-representations closely resemble them [60, 64]. Studies have shown that using self-similar representation in a virtual environment can lead to more original ideas and enhance motivation and performance [34, 42]. Overall, these results suggest that the appearance of people's self-representation can change their subjective experience and performance, and shape how they interact with others.

**2.1.2 Other Users' Avatars' Appearance.** Previous studies have shown that the appearance of other users' avatars can influence perceptions and communication with them. Research has highlighted how various aspects of appearance, such as realism, anthropomorphism, gender, and race, can enhance or discourage interaction [45, 47, 51, 63]. For example, studies have found that people are more likely to continue a conversation with an avatar that is more elaborate or physically attractive [7]. Other research has explored how perceptions of other users' avatars' appearance can affect task load and cognitive performance [33]. Overall, this line of research suggests that the appearance of other users' avatars has a strong effect on people's perceptions, behavior, and task performance. However, the impact of appearance-similarity between self-representations and other users' avatars in social VR is under-explored. This study aims to extend knowledge of the visible characteristic, overall appearance, by investigating how the relationship between self-representations' appearance and other users' avatars can influence the persuasiveness of statements made by the latter.

### 2.2 Similarity between Self-Representations' Appearance and that of Other Users' Avatars

The similarity effect means that people tend to prefer and be influenced more by others who are similar to themselves than dissimilar [11, 46]. Previous studies have shown that similarity is an important factor in attraction [11], intimacy [24], and social judgment [37]; specially, the similarity effect of physical appearance has been studied in contexts [5, 14] such as decision games [14, 15, 17], persuasion [50], and cooperation [21, 35]. The similarity effect of avatar appearance has also been reported in traditional screen-based virtual environments [48, 50, 56], where people tend to recognize each other through salient features like appearance. As a result, appearance similarity between avatars can impact perception, identification, social influence, and task performance [6, 10, 48]. This relationship between similarity of avatar appearance and social influence is likely due to people feeling more fluent and able to form group identification when interacting with avatars similar to their own [10, 29].

Recently, Van Der Land et al. [56] examined the influence of self-similar representation and team-similar other users' avatars in the text-based chat application. They found that when a participant and a group of other users all closely resembled the same avatar, it induced the highest levels of social attraction and markedly enhanced their collaboration performance. However, those findings were arrived at based on morphing-photograph data. Thus, the similarity effect of the avatar's appearance impact in a more embodied

VR environment, in which the digital representations are created by users themselves, remains unknown.

## 2.3 Research Questions

The various research efforts reviewed above suggest that the appearance of both self-representations and other users' avatars has a range of potentially important impacts. However, published data about such impacts within social VR – particularly on persuasiveness – remain scant. Therefore, we will seek to answer the following research questions.

**RQ1** - How do social VR users perceive and communicate with other users' avatars whose appearance is similar to their self-representations?

**RQ2** - Do different levels of appearance similarity between social VR users' self-representations and other users' avatars influence the latter's persuasiveness?

## 3 METHOD

### 3.1 Self-Representation and Avatar Creation

Our study used Meta Avatar System software to create avatars. The interface allows users to design self-representations based on physical characteristics, facial features, and outfits. Facial features were classified into three categories [1, 44], such as face shapes (oval, diamond, round), eyes (downturned, upturned, round) and so on. The classification system also helped to manipulate the appearance-similarity levels. Participants were instructed to create self-representations based on their physical appearance [19], without giving clear instructions, participants may create a self-representation avatar by referring to various factors (Ex. idealized self-images and individual gender identities [19, 20, 34, 48]. The avatar-creation phase of the experiment usually took 15-20 minutes.

### 3.2 The Social Virtual Reality Environment and Apparatus

The experiment presented in this article was conducted in Horizon Workrooms V1.3. The experimental setup for each participant consisted of an Oculus Quest 2 headset, two controllers, Vive Tracker and earphones. To distract the participants' attention from the fact that, in all three conditions, the other avatar was controlled by the same other person, we manipulated that other person's voice using Clownfish Voice Changer [38, 59]. We manipulated the voice pitch in per appearing condition.

### 3.3 Participants

Through a Facebook group at the nearby university, we recruited 25 native Chinese-speaking adults (17 females, Mean age = 23.12, SD = 3.2) to participate in this study. All participants were compensated \$20 for the two hours study. We eliminated two participants' data because of technical issues.

### 3.4 Study Design

The experiment had a within-subjects design, in which each participant performed one discussion task in each of three experimental conditions, which differed in terms of their avatar-interlocutor's

appearance similarity. According to previous research on face recognition and the discrimination of human facial features [28, 41, 41], we manipulated similarity by changing external and internal facial features in the identity, moderate similarity, and dissimilarity conditions. In the moderate similarity condition, we only changed one external feature of the avatar from the participants' initially selected category to another category, with all other facial features remaining the same. In the dissimilarity condition, we changed two external features (hair, face shape) and four internal ones (eye, eyebrow, nose, and mouth shapes) to create a less similar appearance. Specifically, the category of face shape would be altered again. The manipulation among three conditions was designed to ensure similar magnitude between each condition for all participants [44]. Participants could see their appearance of self-representation through the mirror in the VR environment, which allowed them to compare the appearance similarity of the researcher-controlled avatars'.

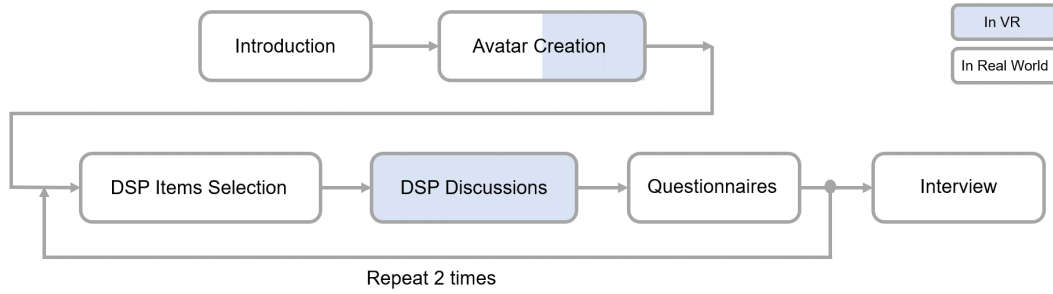
To evaluate the influence of appearance similarity on communication outcome in each condition, we used a modified version of DSPs [36, 54, 59], which measures persuasiveness in conversation. We expanded the number of item pairs in the modified DSP based on prior research [52] and our pilot tests. Participants chose one item from five pairs based on which they felt was most important for survival in a desert in each condition. They then discussed their choices with another "participant" (researcher-controlled avatar) and were asked if they wanted to change their choices, with persuasiveness measured by counting the number of item pairs in which the participants changed their choices. The specific items in the pairs and choice-mismatches varied across conditions to ensure that the participant would not suspect that all their interlocutors were controlled by the same person [54, 59]. On average, the conversation took about 9 minutes, irrespective of experimental conditions.

### 3.5 Procedure

In Figure 2, the procedure of the experiment is illustrated. The participants were first given a dummy purpose, which was to test how people experience communication in a VR environment as avatars. They then created their self-representations and were instructed to replicate their physical appearance as closely as possible. After they left the VR environment temporarily, they were given a form describing the DSP scenario and five item pairs. They were also given rules for their avatar-to-avatar conversation, with the goal of discussing their item choices. They then returned to the VR environment and had a discussion with the researcher-controlled avatar while seeing their self-representation in a mirror. After the discussion, they submitted their final item choices and completed questionnaires about the avatar they interacted with. At the end of the experiment, we conducted a semi-structured interview. The entire experiment, including the introduction, avatar creation, three discussions, short breaks, and an interview, lasted 90-100 minutes.

### 3.6 Data Collection and Analysis

**3.6.1 Quantitative Data.** In all three conditions, the following quantitative measurements were employed to allow us to analyze the



**Figure 2: The experimental procedure, illustrated in figure 2, shows the VR and real-world phases.**

influence of the appearance-similarity of other users' avatars on the participants.

(1) *Avatar Similarity*. Our main independent variable (IV) was the similarities in appearance of other avatars to the participant's self-representation. To check our manipulation of appearance was successful or not, we applied the avatar similarity questionnaire [34, 58] per conditions per participants. Participants rated how similar they felt the other avatar was to their self-representations in an appearance on a seven-point Likert scale (1 = not at all, 4 = neutral, 7 = completely ).

(2) *Persuasiveness*. To evaluate the influence of appearance similarities on communication outcomes, we computed the total number of desert-survival items about which the participants changed their initial choices. Because the researcher-controlled avatar only disagreed about four out of the five item choices per condition, the possible score range was 0-4 in a given condition. This is a common metric for persuasiveness used in previous DSP studies [52, 54, 59].

(3) *Uncanny Valley*. The Uncanny Valley questionnaire measured the strength of the participants' feelings of unpleasantness or discomfort triggered by our experimental manipulations of appearance and/or avatars' abnormal body movements [16]. It relies on 19 semantic-differential items, responded to on a five-point bipolar response format. The items represent three dimensions: perceived humanness, attractiveness, and eeriness.

**3.6.2 Qualitative Data.** We recorded and transcribed what the participants said in their three DSP discussions and in their interviews. Each of these two categories of data is described in turn below.

(1) *Discussion Content*. This study analyzed the discussion transcripts by using the Linguistic Inquiry and Word Count tool [27, 49] to calculate the percentage of words used by the participants in several categories. These categories were chosen as they reflect the participant's feelings, focus, and thoughts [55]. For instance, pronouns can reveal the social distance between interlocutors and the relationship quality [18, 26, 55]. The categories used in this study were pronouns, affective processes, cognitive processes, and social processes to evaluate the general feelings and perceptions of the conversation in each condition.

(2) *Interview Data*. In the semi-structured interviews we conducted at the end of the experiment, we mainly focused on the participants' general feelings about or impressions of conversation in each condition; whether they had noticed that some other users' avatars looked similar to their self-representations; which of

the three other users' avatars they preferred communicating with; and why they preferred to interact with that avatar. The average interview duration was 20 minutes.

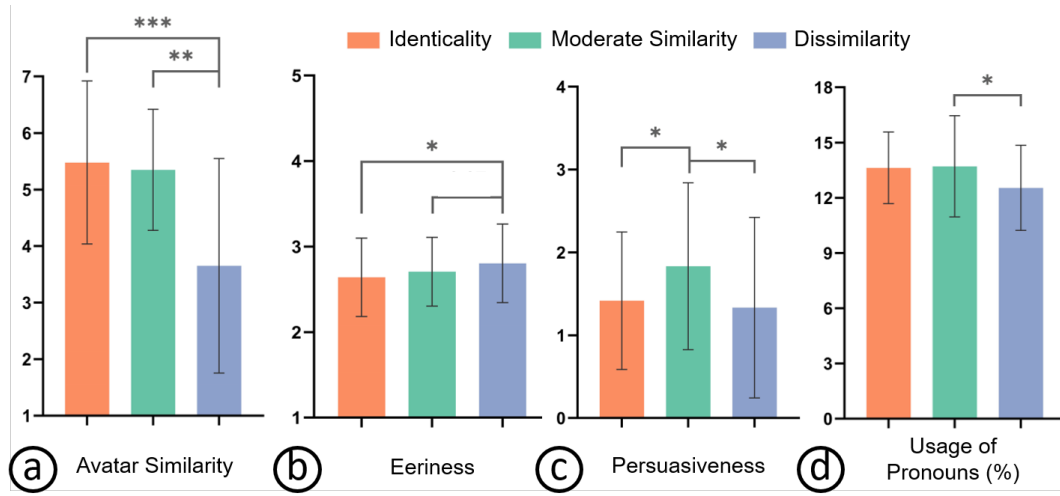
**3.6.3 Analysis.** The quantitative data and the discussion-content component of the qualitative data were analyzed using one-way repeated measures analyses of variance (ANOVAs). Post hoc tests used Tukey's HSD with  $\alpha = 0.05$ . All data were assessed using Mauchly's test for sphericity, and Greenhouse-Geisser corrections were applied as needed. We also report the effect size using the partial eta-squared  $\eta_p^2$  as an estimate, and classify it according to Cohen's convention [13]. For the interview data, we used thematic content analysis [43, 53], which involves an iterative process to label participants' responses with emerging themes. The process focused on identifying recurrent themes or concepts representing the participants' feelings about, impressions of, or preferences of each condition [19].

## 4 RESULTS

### 4.1 Manipulation Check and Uncanny Valley

We commenced our analysis by examining the effectiveness of avatar appearance manipulation, as reflected in the post-interaction questionnaire data. The results revealed a significant main effect of avatar similarity on similarity perceptions ( $F(2, 42) = 9.310, p < .001, \eta_p^2 = .307$ ) with a large effect size. Post hoc tests confirmed that the participants were significantly more likely to think the other user's avatar's appearance was similar to their appearance of self-representation in the identity condition ( $M = 5.47, SD = 1.44, p < .001$ ) and moderate similarity ( $M = 5.34, SD = 1.07, p < .01$ ) condition than in the dissimilarity condition ( $M = 3.62, SD = 1.89$ ). There was no significant difference between the identity and moderate similarity conditions ( $p = .91$ , Figure 3a).

The main effects of appearance similarity on two of the three Uncanny Valley categories, perceived humanness and attractiveness, were non-significant: respectively ( $F(2, 42) = 1.42, p = .25$  and  $F(2, 42) = 0.79, p = .46$ , respectively). However, there was a significant main effect of appearance similarity on Uncanny Valley's other component, eeriness ( $F(2, 42) = 4.196, p = .02, \eta_p^2 = .022$ ) with a large effect size. Post hoc tests confirmed that participants felt the other users' avatars were significantly eerier in the dissimilarity condition ( $M = 2.80, SD = 0.46, p = .02$ ) than in the identity condition ( $M = 2.64, SD = 0.46$ ). Also, participants tend



**Figure 3: (a) Manipulation Check, showing participant ratings of avatar similarity in each condition. (b) Uncanny Valley ratings of attractiveness, perceived humanness, and eeriness in each condition. (c) Persuasiveness effects of different levels of avatar similarity on number of desert-survival items changed from the participants’ initial choices. (d) Percentages of pronouns used in each condition. The error bars denote standard deviation, and \*, \*\*, and \*\*\* significant differences at  $p < 0.05$ ,  $p < 0.01$ ,  $p < 0.001$  levels, respectively.**

to feel other users’ avatars are eerier in the dissimilarity condition than in the moderate similarity condition ( $M = 2.70$ ,  $SD = 0.40$ ,  $p = .07$ ), but there was no significant difference between the identity and moderate similarity conditions ( $p = .89$ ; see Figure 3b).

## 4.2 Persuasiveness and Discussion Content

Figure 3c shows the numbers of desert-survival items that the participants changed their initial choices of after interacting with a researcher-controlled avatar. The results suggest that there was a significant main effect of avatar appearance similarity on persuasiveness ( $F(2, 42) = 4.72$ ,  $p = .01$ ,  $\eta_p^2 = .184$ ) with a large effect size. Post hoc tests confirmed that participants changed their initial choice most often after discussion with an avatar of moderate similarity ( $M = 1.87$ ,  $SD = 1.01$ ) than with either an identical one ( $M = 1.39$ ,  $SD = 0.84$ ,  $p = .02$ ) or a dissimilar one ( $M = 1.35$ ,  $SD = 1.11$ ,  $p = .04$ ). The difference between the identity and dissimilarity conditions in this respect was non-significant ( $p = .95$ ).

One of four categories varied depending on the conditions (see Figure 3d), revealing that there was a significant main effect of avatar appearance similarity on the usage of pronouns ( $F(2, 42) = 3.69$ ,  $p = .03$ ,  $\eta_p^2 = .146$ ) with a large effect size. Post hoc testing confirmed that the participants significantly used more pronouns during discussions with moderately similar avatars ( $M = 13.71$ ,  $SD = 2.75$ ) than with dissimilar ones ( $M = 2.32$ ,  $SD = 0.48$ ,  $p = .04$ ). Participants tend to use more pronouns during the discussion in the identity condition than in the dissimilarity condition ( $M = 13.63$ ,  $SD = 0.41$ ,  $p = .07$ ). However, there was no significant difference between identity and moderate similarity conditions ( $p = .98$ ).

## 4.3 Qualitative Feedback

**4.3.1 Intimacy with a Similar Other as a Spur to Conversation.** The participants expressed generally positive feelings about conversing with avatars who had a similar appearance to their self-representations. Many ( $n=8$ ) associated such feelings with intimacy, stating they tended to pay more attention when communicating with an avatar of similar appearance. P11, for instance, commented: “when seeing an avatar whose appearance is similar to mine, I feel more intimate with him, and thus that it’s easier to communicate; conversely, I feel distance between myself and a dissimilar avatar.” Such sentiments were echoed by three other participants (P1, P7, P18), who commented that experience of both identity and moderate similarity conditions had been like talking to themselves in front of a mirror.

**4.3.2 Possible Confusion Caused by Identity.** However, identical avatars seemed to obstruct various participants’ sense of agency ( $n=8$ ). P12 said: “It’s weird the avatar is the same-looking but is not controlled by me.” One participant (P22) shared a notable confusion and told us that seeing the identical other led her to establish a belief or expectation that it would exhibit similar opinions about the DSP questions and/or similar personality traits to her own. This expectation was soon dashed after starting a conversation, however, and this cooled the interaction.

## 5 DISCUSSION

### 5.1 The Influence of Appearance Similarity on Interaction in Social Virtual Reality

**5.1.1 Perceptions and Communication Patterns.** Our first key finding is that people appear to differentially perceive, and communicate differently with, avatars of similar vs. dissimilar appearance to

their self-representations **RQ1**. Participants felt more intimate and fluent when communicating with avatars that were similar to their self-representations, as evidenced by increased use of pronouns [26, 57] and positive feedback in interviews. This aligns with previous research that has shown that when interlocutors share similar facial appearance, it can trigger positive reactions such as attraction, trust, and fluency [12, 14, 15, 17, 56]. However, the results on eeriness were mixed. Based on previous research [47], we suspect that the overall eeriness ratings were lower in the identity condition and moderate similarity conditions than in the dissimilarity condition because it was compensated for by intimacy or comfort, which in the latter condition was rarely felt.

**5.1.2 Influence of Persuasiveness for Communicating with a Similar Appearance Avatar.** Regarding **RQ2**, our results indicate that an avatar of similar appearance – especially moderate similarity – will have measurably greater persuasiveness than a dissimilar one. This broadly echoes the findings of previous studies in which people preferred and tend to be influenced more by similar others than dissimilar ones [11, 46]. The present results confirm similar findings (i.e., positive influence of appearance similarity [12, 14, 22, 35, 50, 56, 56]) in previous research and extend them to the context of social VR. Nevertheless, our participants were not more persuaded in the identity condition than in the moderate similarity one. This may be because the identical-appearance avatar caused confusion about agency, leading to negative perceptions that negatively impacted persuasiveness. Previous studies have reported corresponding findings, with participants feeling uncomfortable or anxious when interacting with identical agents [23] due to mismatches between identification and interactivity [3, 4, 30]. Therefore, it is reasonable to suppose that because of such a mismatch, our own participants' sense of the persuasiveness of researcher-controlled avatars in the identity condition would have been low.

## 6 CONCLUSION

This paper has addressed how the similarity effect impacts people interacting via embodied avatars in an immersive social VR environment. Our key findings are 1) that avatars of moderate appearance similarity had the greatest persuasiveness; 2) that conversing with an avatar of either identical or moderately similar appearance triggered positive feelings such as intimacy and comfort, which in turn tended to boost conversational fluency and compensate for the other avatar's eeriness. We hope that these contributions to the HCI community will inform future theoretical and practical work aimed at creating more meaningful and thought-provoking avatar-based social VR environments.

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